Noisy U-TV | NPL-default vs no noise

Frozen best checkpoint: epoch 5, step 22,500. No retraining.

6 ratios: 5, 10, 20, 30, 40, 50%; multilevel coarse J=6.

50 held-out test images: 10 per category; 300 reconstructions per condition.

FIXED NPL: existing PCMNoiseSimulation.npl_default preset, unmodified.

NO NOISE: noise=None in LION acquisition simulator.

Identical sampling masks for paired NPL and clean tests at each image and ratio.

One NPL noise realisation per image/ratio; no noise replicates.

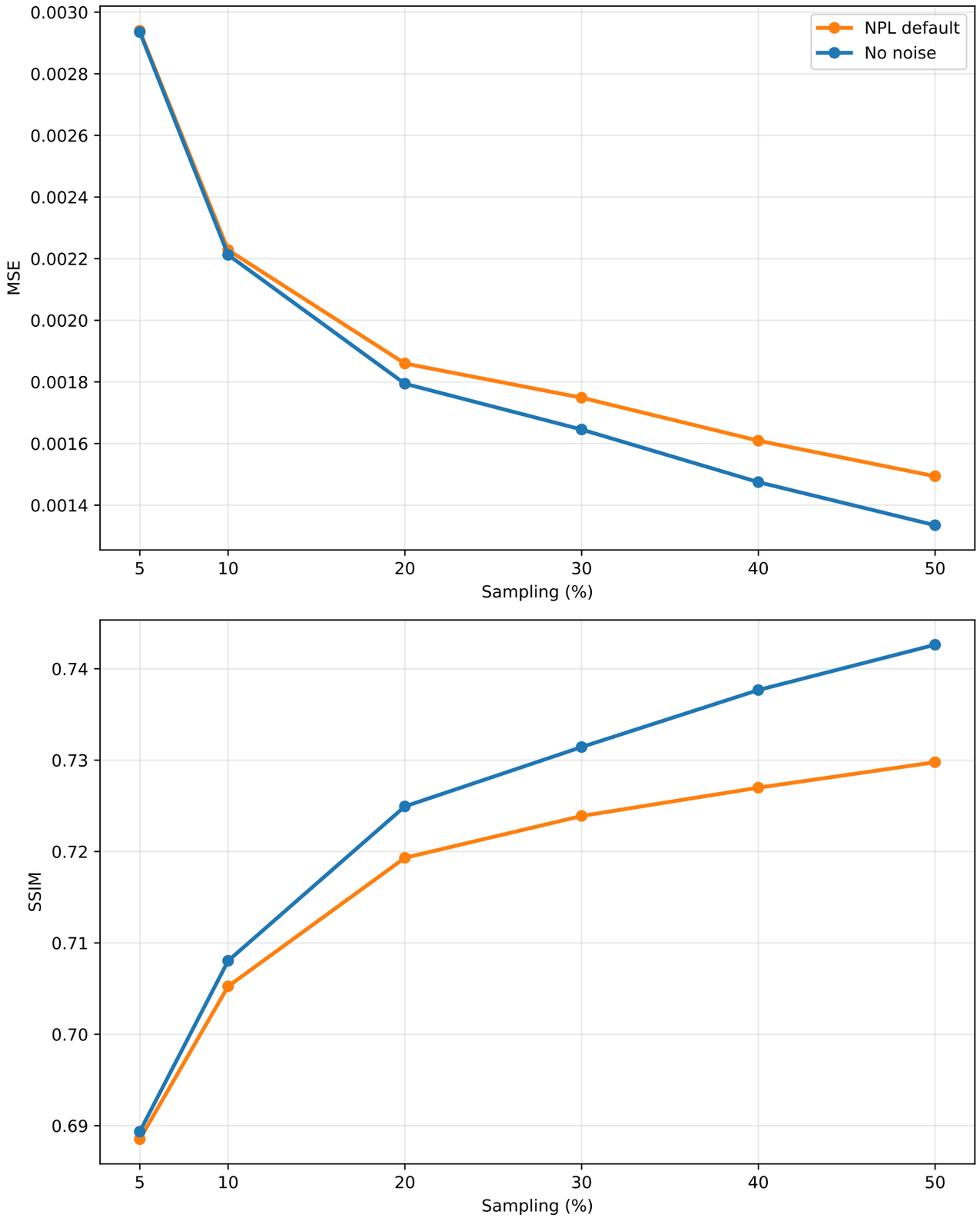
All reconstructions use one shared parameter-map scale factor key.

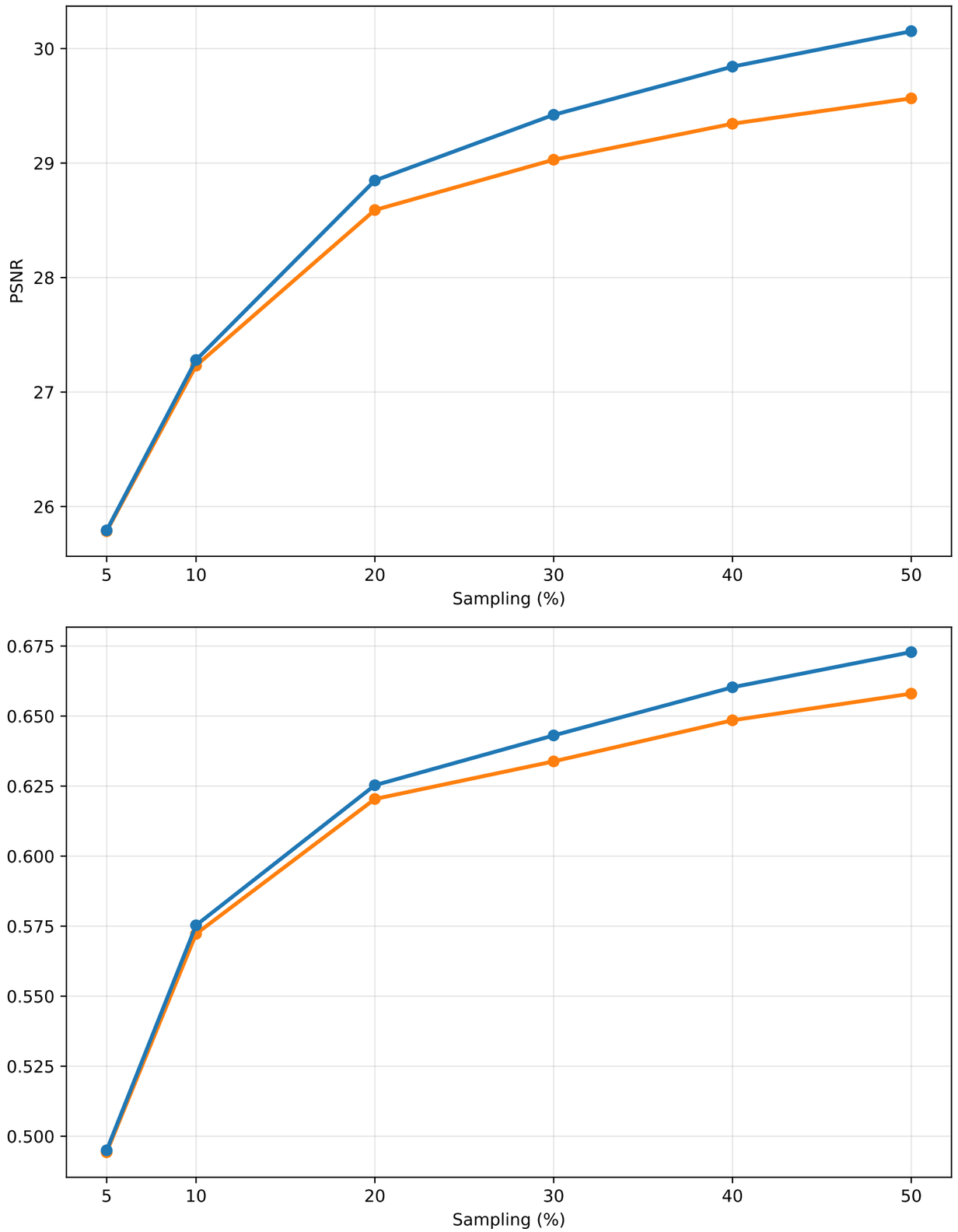
Reconstruction plots use existing LION renderer, fixed model scale [0, 1].

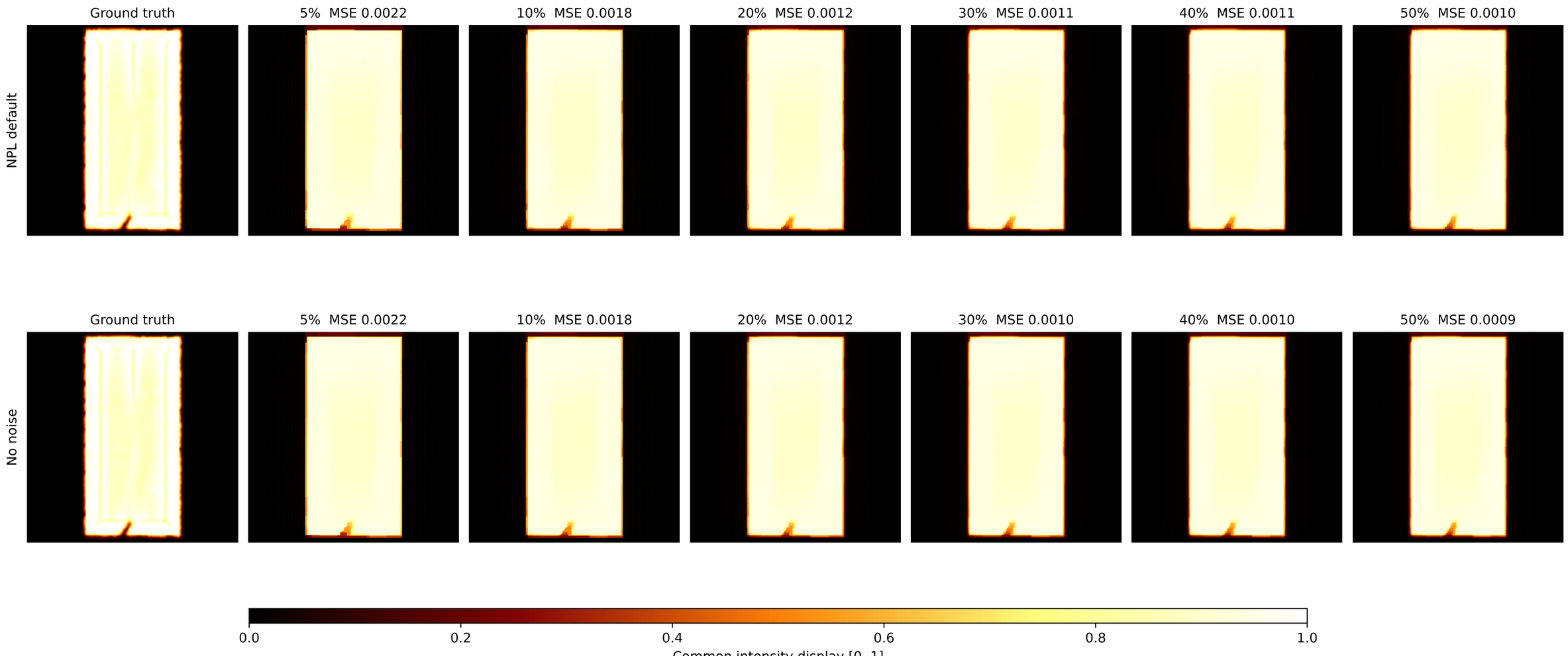
The previously reported variable-noise sweep is NOT included in these averages.

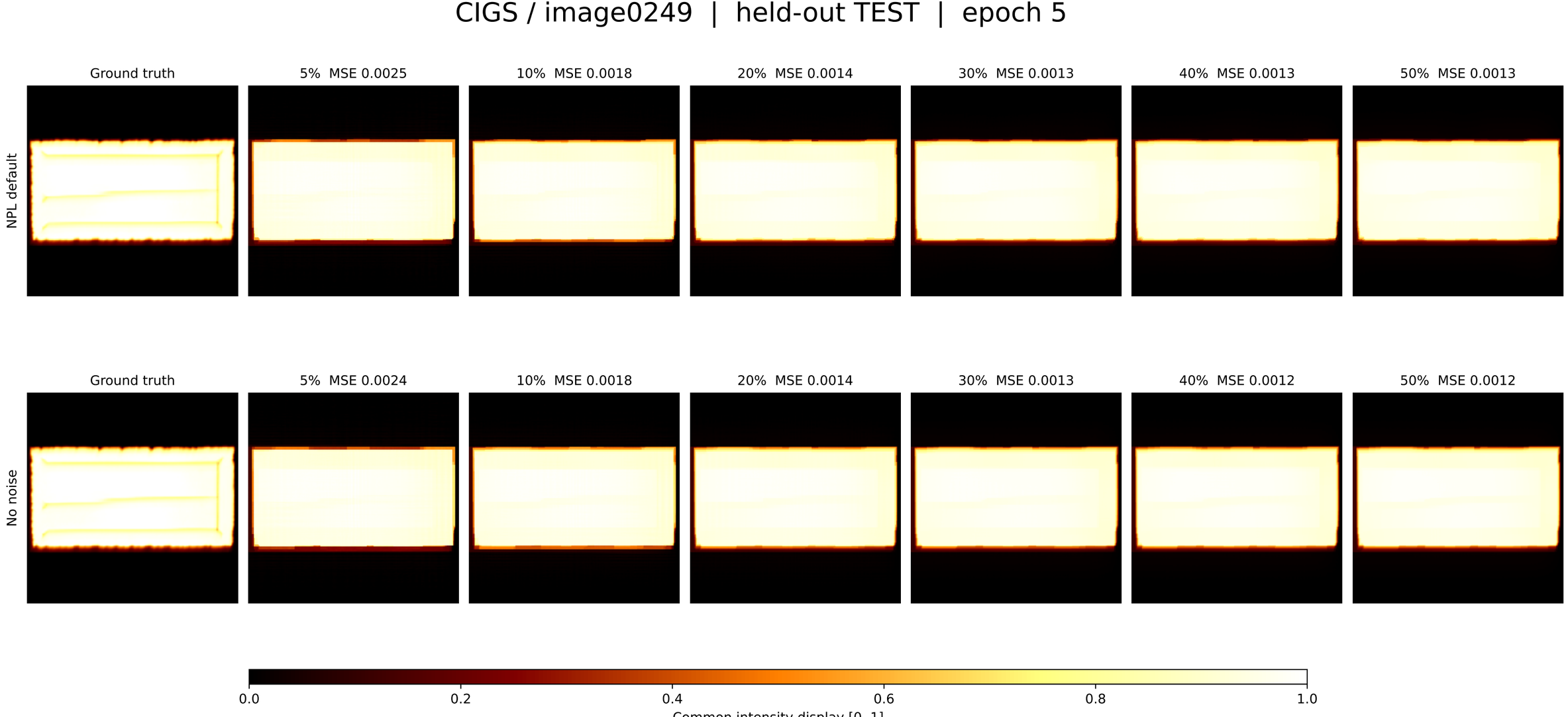
Sampling	NPL MSE	Clean MSE	NPL PSNR	Clean PSNR
5%	0.00294	0.00294	25.78	25.79
10%	0.00223	0.00221	27.23	27.28
20%	0.00186	0.00179	28.59	28.85
30%	0.00175	0.00165	29.03	29.42
40%	0.00161	0.00147	29.34	29.84
50%	0.00149	0.00133	29.57	30.15

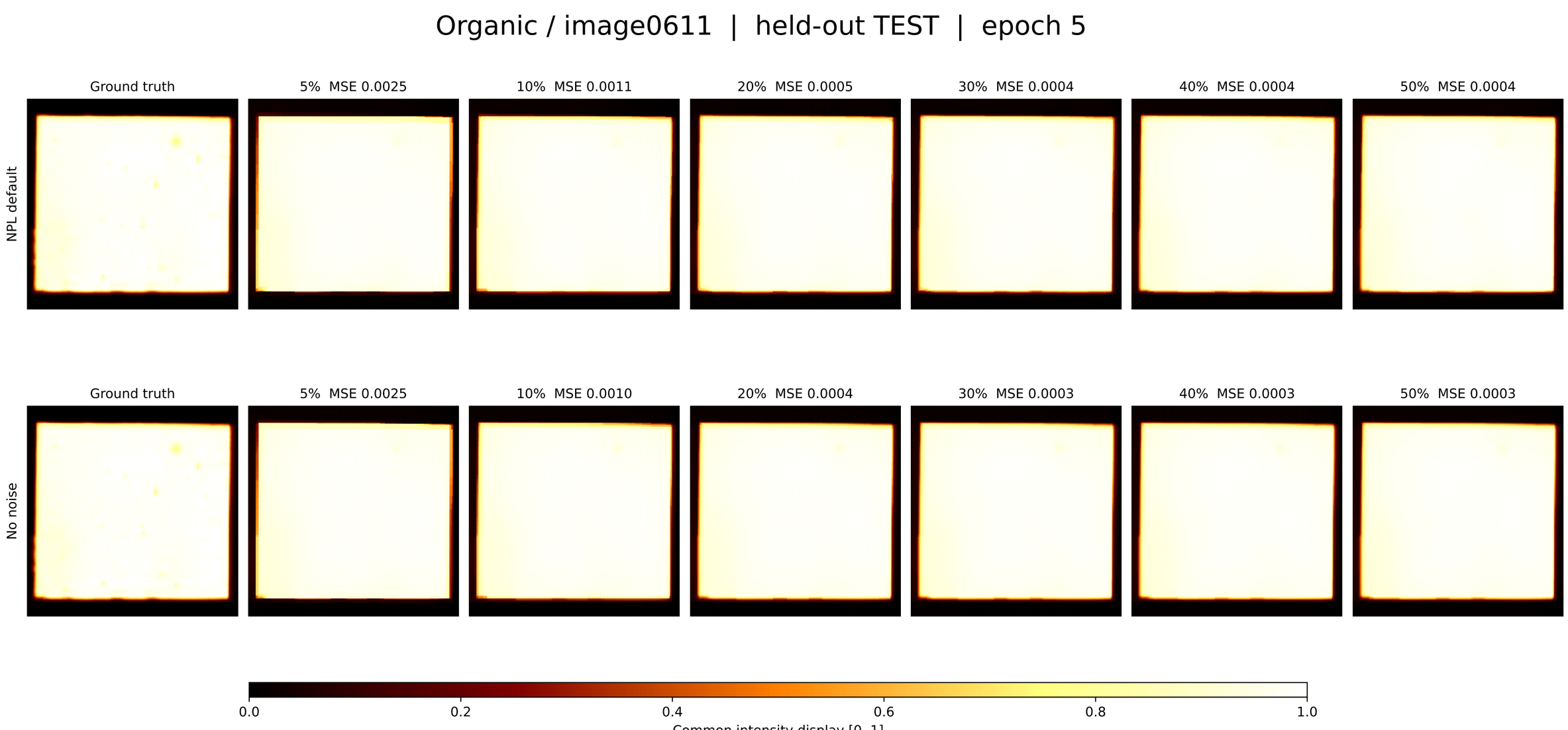
Controlled acquisition comparison (same images and masks)

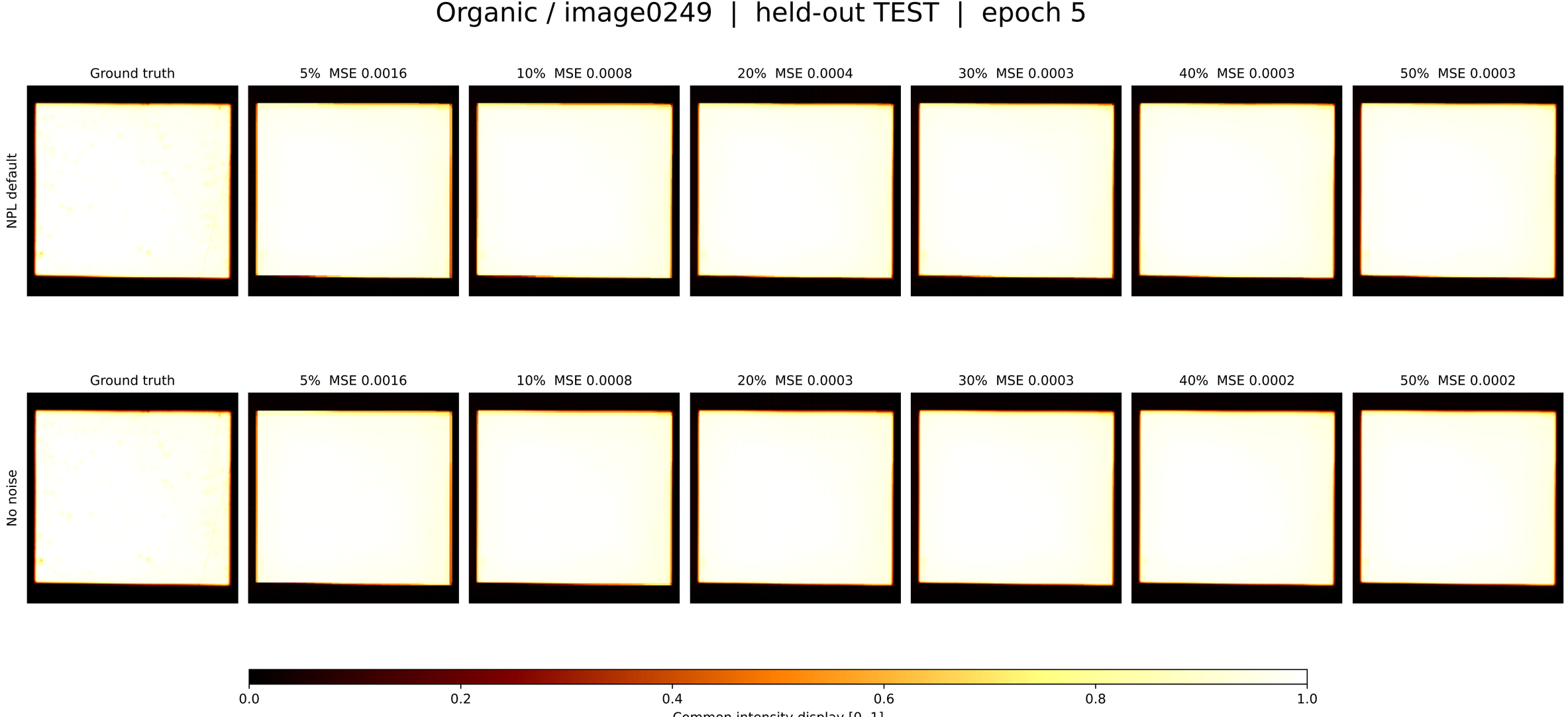


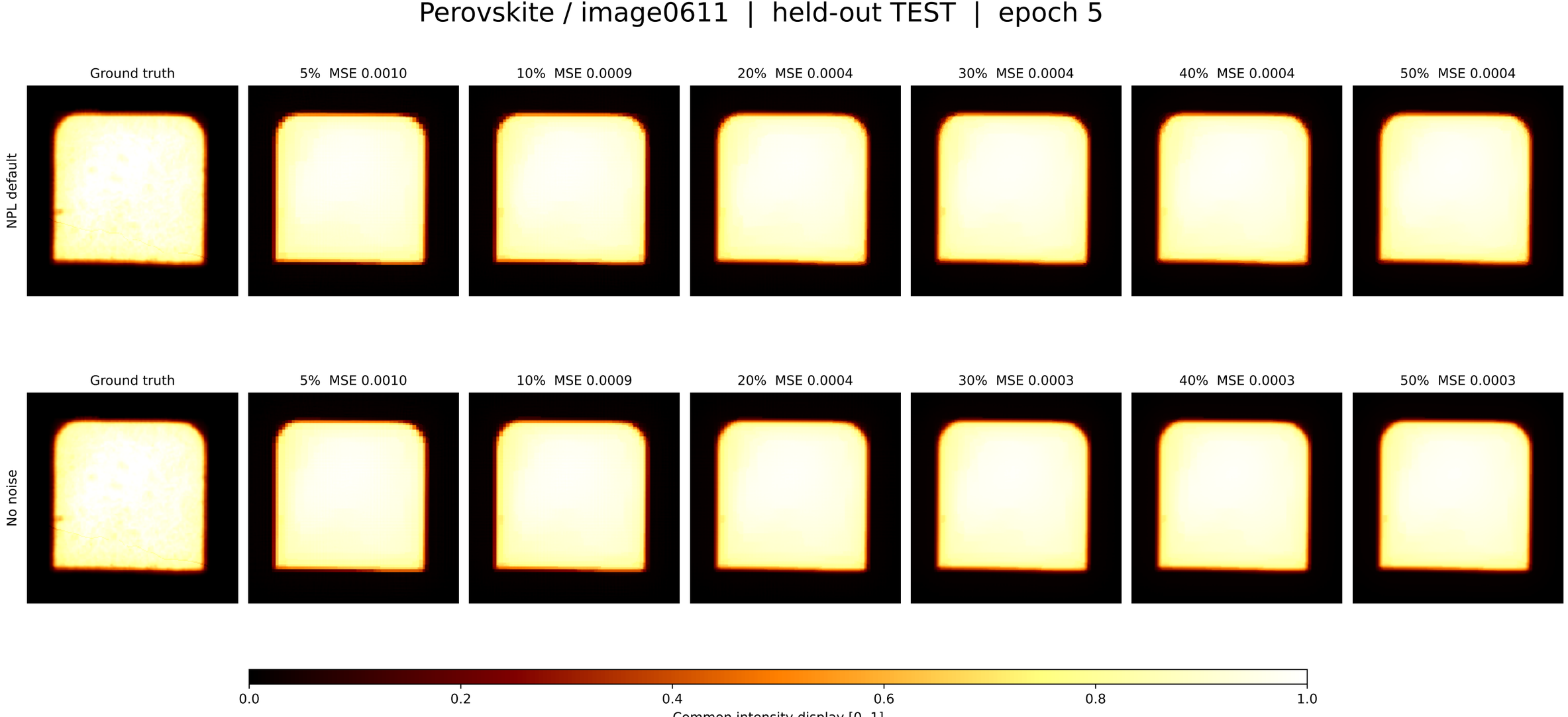


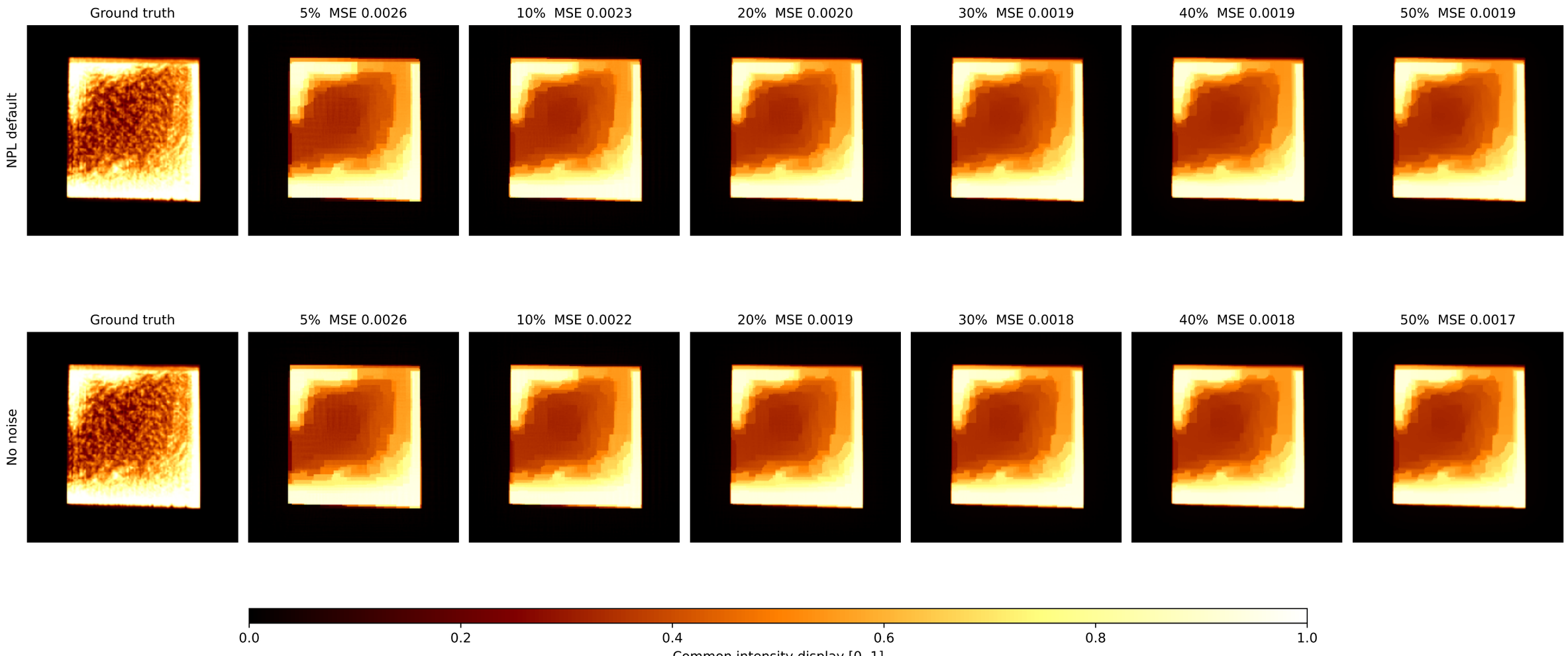

CIGS / image0611 | held-out TEST | epoch 5
NPL default
Ground truth
5% MSE 0.0022
10% MSE 0.0018
20% MSE 0.0012
30% MSE 0.0011
40% MSE 0.0011
50% MSE 0.0010
No noise
Ground truth
5% MSE 0.0022
10% MSE 0.0018
20% MSE 0.0012
30% MSE 0.0010
40% MSE 0.0010
50% MSE 0.0009
0.0
0.2
0.4
0.6
0.8
1.0

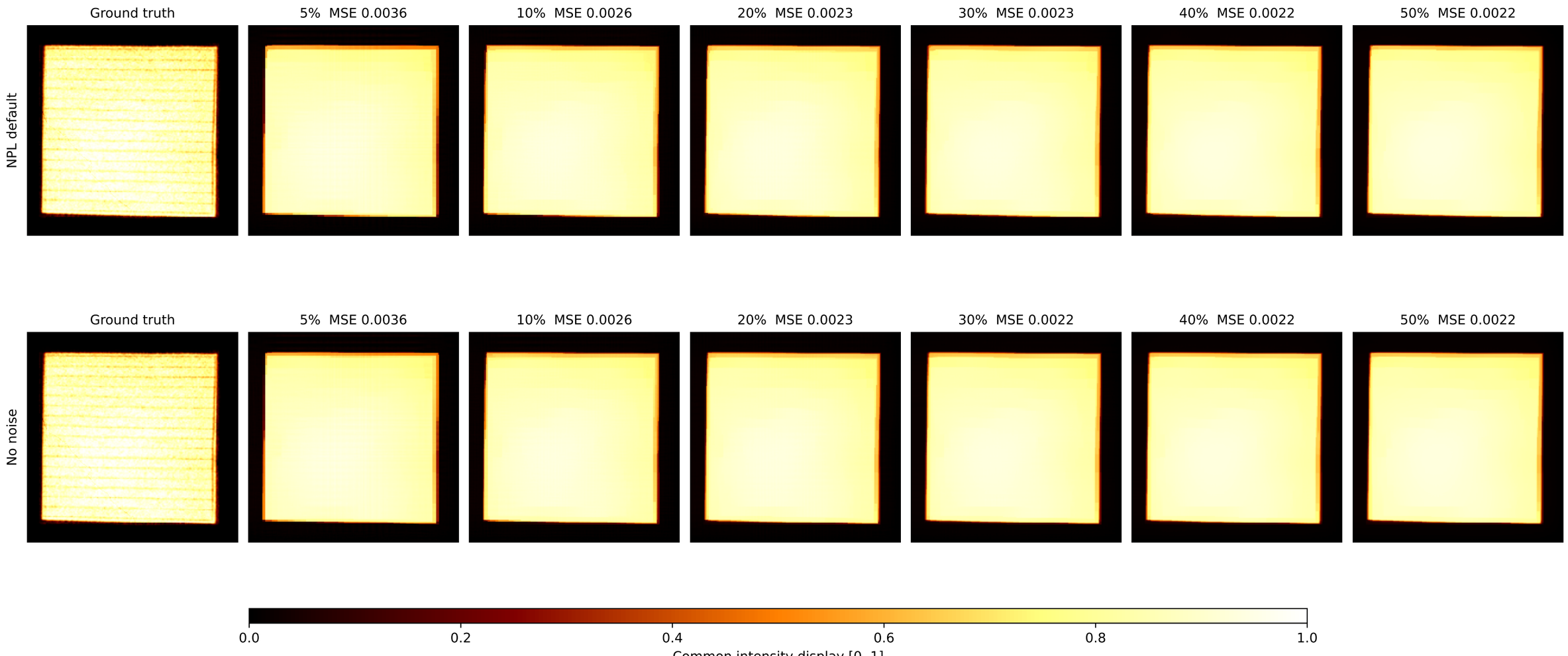

CIGS / image0249 | held-out TEST | epoch 5
Ground truth
5% MSE 0.0025
10% MSE 0.0018
20% MSE 0.0014
30% MSE 0.0013
40% MSE 0.0013
50% MSE 0.0013
NPL default
Ground truth
5% MSE 0.0024
10% MSE 0.0018
20% MSE 0.0014
30% MSE 0.0013
40% MSE 0.0012
50% MSE 0.0012
No noise
0.0
0.2
0.4
0.6
0.8
1.0

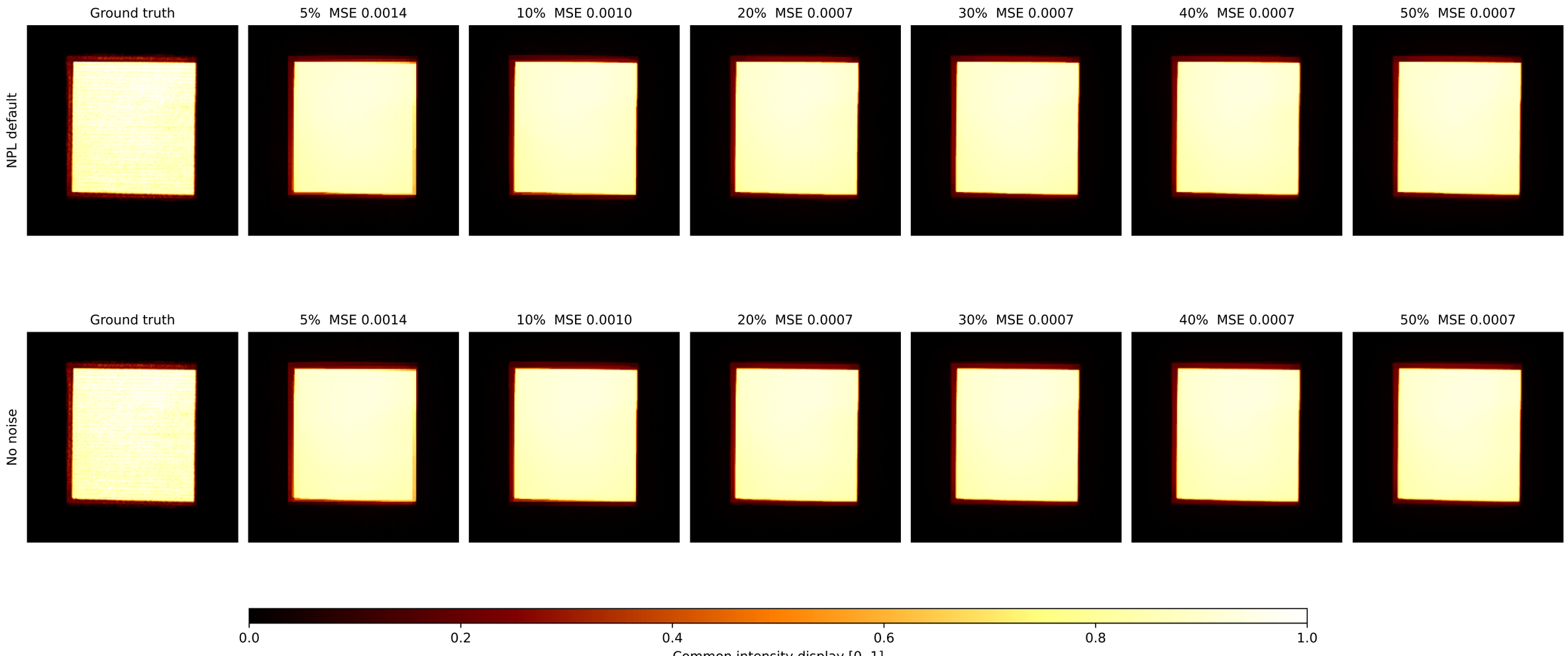

Organic / image0611 | held-out TEST | epoch 5
Ground truth
5% MSE 0.0025
10% MSE 0.0011
20% MSE 0.0005
30% MSE 0.0004
40% MSE 0.0004
50% MSE 0.0004
NPL default
Ground truth
5% MSE 0.0025
10% MSE 0.0010
20% MSE 0.0004
30% MSE 0.0003
40% MSE 0.0003
50% MSE 0.0003
No noise
0.0
0.2
0.4
0.6
0.8
1.0
Common intensity display [0, 1]

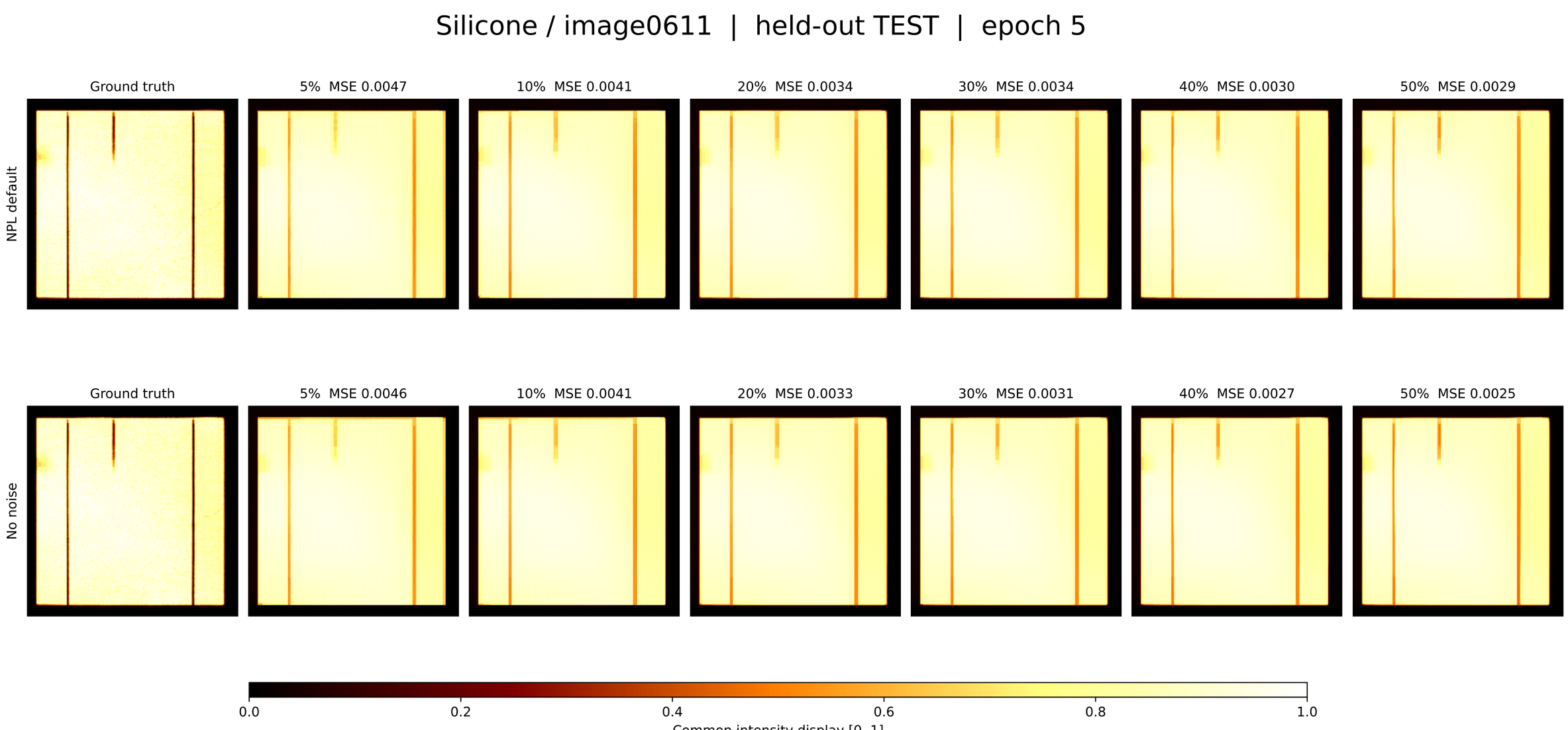

Organic / image0249 | held-out TEST | epoch 5
Ground truth
5% MSE 0.0016
10% MSE 0.0008
20% MSE 0.0004
30% MSE 0.0003
40% MSE 0.0003
50% MSE 0.0003
NPL default
Ground truth
5% MSE 0.0016
10% MSE 0.0008
20% MSE 0.0003
30% MSE 0.0003
40% MSE 0.0002
50% MSE 0.0002
No noise
0.0
0.2
0.4
0.6
0.8
1.0

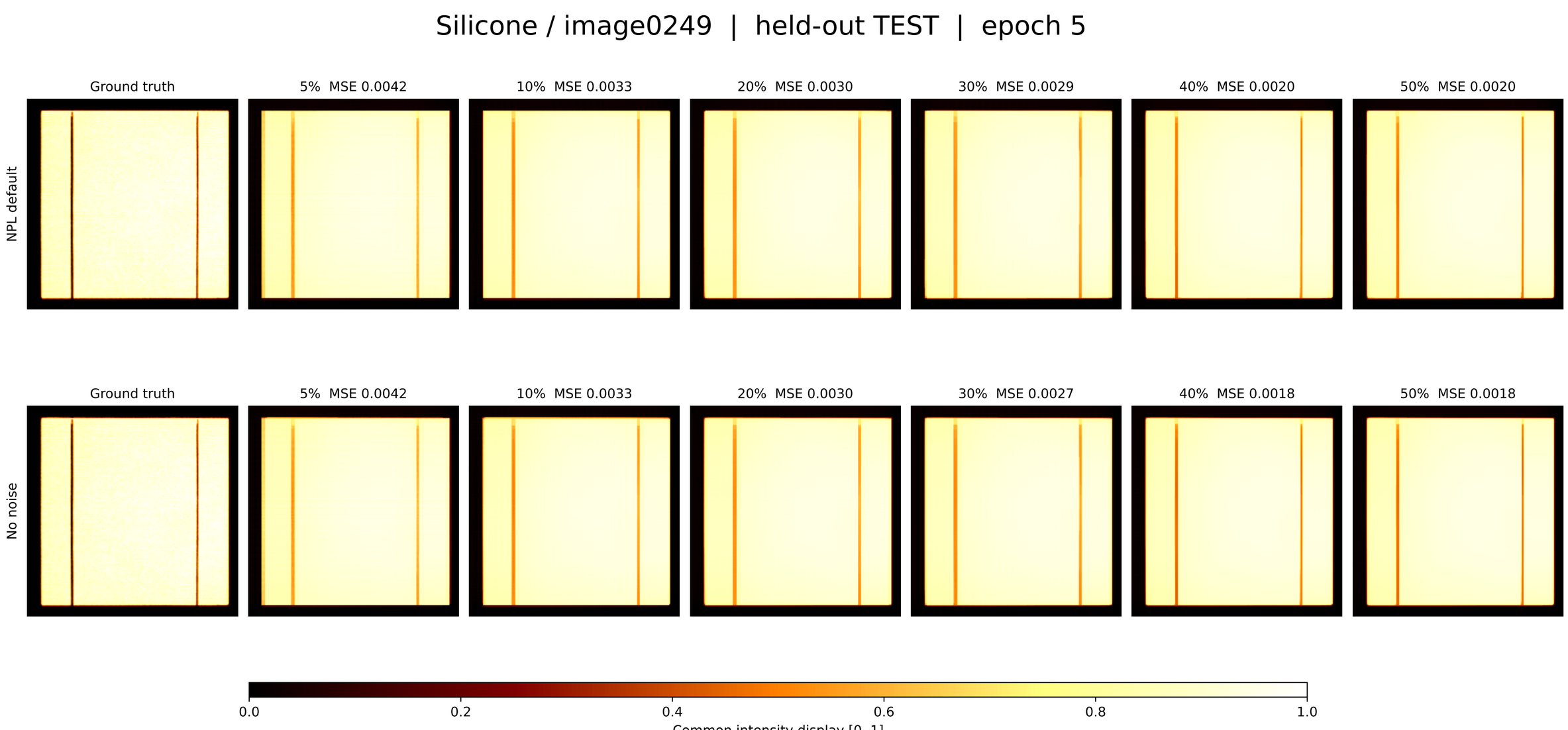

Perovskite / image0611 | held-out TEST | epoch 5
NPL default
Ground truth
5% MSE 0.0010
10% MSE 0.0009
20% MSE 0.0004
30% MSE 0.0004
40% MSE 0.0004
50% MSE 0.0004
No noise
Ground truth
5% MSE 0.0010
10% MSE 0.0009
20% MSE 0.0004
30% MSE 0.0003
40% MSE 0.0003
50% MSE 0.0003
0.0
0.2
0.4
0.6
0.8
1.0


Perovskite / image0249 | held-out TEST | epoch 5
Ground truth
5% MSE 0.0026
10% MSE 0.0023
20% MSE 0.0020
30% MSE 0.0019
40% MSE 0.0019
50% MSE 0.0019
NPL default
Ground truth
5% MSE 0.0026
10% MSE 0.0022
20% MSE 0.0019
30% MSE 0.0018
40% MSE 0.0018
50% MSE 0.0017
No noise
0.0
0.2
0.4
0.6
0.8
1.0


Reference / image0611 | held-out TEST | epoch 5
NPL default
Ground truth
5% MSE 0.0036
10% MSE 0.0026
20% MSE 0.0023
30% MSE 0.0023
40% MSE 0.0022
50% MSE 0.0022
No noise
Ground truth
5% MSE 0.0036
10% MSE 0.0026
20% MSE 0.0023
30% MSE 0.0022
40% MSE 0.0022
50% MSE 0.0022
0.0
0.2
0.4
0.6
0.8
1.0


Reference / image0249 | held-out TEST | epoch 5
Ground truth
5% MSE 0.0014
10% MSE 0.0010
20% MSE 0.0007
30% MSE 0.0007
40% MSE 0.0007
50% MSE 0.0007
NPL default
Ground truth
5% MSE 0.0014
10% MSE 0.0010
20% MSE 0.0007
30% MSE 0.0007
40% MSE 0.0007
50% MSE 0.0007
No noise
0.0
0.2
0.4
0.6
0.8
1.0


Silicone / image0611 | held-out TEST | epoch 5
NPL default
Ground truth
5% MSE 0.0047
10% MSE 0.0041
20% MSE 0.0034
30% MSE 0.0034
40% MSE 0.0030
50% MSE 0.0029
No noise
Ground truth
5% MSE 0.0046
10% MSE 0.0041
20% MSE 0.0033
30% MSE 0.0031
40% MSE 0.0027
50% MSE 0.0025
0.0
0.2
0.4
0.6
0.8
1.0


Silicone / image0249 | held-out TEST | epoch 5
NPL default
Ground truth
5% MSE 0.0042
10% MSE 0.0033
20% MSE 0.0030
30% MSE 0.0029
40% MSE 0.0020
50% MSE 0.0020
No noise
Ground truth
5% MSE 0.0042
10% MSE 0.0033
20% MSE 0.0030
30% MSE 0.0027
40% MSE 0.0018
50% MSE 0.0018
0.0
0.2
0.4
0.6
0.8
1.0